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EDUCATION

University of California, Berkeley Expected 5/2020
Master of Engineering, Bioengineering
Anna University, Chennai, India 6/2019
Bachelor of Engineering, Biomedical Engineering

SOFTWARE SKILLS

C, C++, Matlab, LabVIEW, and Solidworks

EXPERIENCE

Graduate Research Assistant, University of California Berkeley **8/19- 12/19**

- Designed the control system of a handheld device for the analysis of optical characteristics of biomolecules.
- Working on portable characterization techniques like DLS and SPR for analysis of biomolecules.

Research Assistant, Indian Institute of Science, Bangalore **1/19-7/19**

- Fabricated and characterized chemical sensors for the sensing of volatile organic compounds.
- Designed PCB boards for the sensor array.

Research Intern, Defense Research and Development Organization, Bangalore **Winter 2017**

- Performed deposition and characterisation of titanium dioxide on glass substrate using RF Sputtering.
- Specialised with RF Sputtering Unit, UV-Vis Spectrometry, XRD, SEM, EDAX and Photoluminescence.

ENGINEERING PROJECTS

A MEMS device for real time manipulation of cell electroporation using electric tweezers

A novel microelectromechanical device was designed for single cell electroporation using photolithography along with the 3D designing of the PDMS chamber using soft lithography. The control system of the device was designed, experiment and manipulated with LabVIEW. The throughput of the device was determined to be in data per microseconds.

Fabrication of gas sensors for the detection of tuberculosis from the exhaled breath

Functionalized GNP-based sensor and CNT-based sensor were fabricated for the detection of Tuberculosis biomarkers from the breath. Characterization of the fabricated gas sensors was observed using spectrometer, FTIR, and TEM. An accuracy of 93% was expected from the diagnostic system.

Deposition and Characterization of Titanium dioxide thin films on glass substrate using RF Magnetron Sputtering

Different thickness of TiO_2 film samples was deposited on glass substrates by varying different parameters in the RF Sputtering system. The samples were characterized by UV-Vis Spectrometer, SEM, EDAX, XRD, and Photoluminescence to determine the ideal conditions for the deposition of TiO_2 on glass substrate.

A handheld Optical characterisation device using STM32 Microcontroller

A handheld spectroscopy using Image Sensor and STM32 Microcontroller was designed and tested for analysis of biomolecules. The spectral analysis is displayed and manipulated in Matlab in microseconds with accuracy of 80%.

Automatic Vehicle for the physically challenged

A cost-effective, locomotive, real-time device was designed for the easy movement of the physically challenged. The control circuit was designed and prototyped for the battery- operated electric vehicle using Arduino microcontroller.

ACHIEVEMENTS

- Received funding by Centre of Healthcare Technologies for the project "Development of gas sensors for the detection of Tuberculosis from exhaled breath" – 2018
- Published paper on "**Design and Development of a mobility aid for the physically challenged**" in **International Journal of Applied Engineering Research** by Research India Publications - August 2018